

How to prepare for the exam?

Recommendations

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The exam is set for 2026-05-27, 2-5 PM

The general arrangements

The coding challenge includes:

- 3 problems (not just coding!); 25% each
- document everything in R Markdown 25%
- 3 hours in total
- open-book.

Packages that are available on the computers in the computer class

- all basic R packages
- `tidyverse`
- `rmarkdown`
- `knitr`
- `tinytex`

All tasks can be solved using just these basic libraries.

Getting ready for the exam

Simple data cleaning maneuvers

- Check for NAs;
- Check for duplications;
- Check which values are in each column (`table()`, `unique()`, any other ideas?);
- Convert to the wide format if applicable;
- Arrange and merge data if it looks reasonable;
- Make your data **suitable for your analysis**.

If you see **Check your data carefully**, you should be careful about the data. It may require data cleaning. Also, **comment on your data cleaning process**. It will simplify our task to understand your data cleaning strategy. The same about your analysis – do not just pile all your plots on us. Explain what they mean at least briefly. Make your answer cohesive.

R Markdown

Keep your report simple and neat. Avoid using sophisticated formatting and avoid copying text from external sources. If it contains Unicode symbols, your R Markdown will likely fail to create a PDF output.

Review semester 1 weeks 13 and 15, and semester 2 weeks 1 and 14 materials for more details about how to use R Markdown. A few hints right away:

- Add empty lines between the elements of your document (headers and the subsequent text, different paragraphs, etc). The more lines you add, the less likely you are to get problems. For instance, compare the following two fragments:

Fragment 1:

```
#header of section 1
## Subheader 1
Some text
```${r}
some code
```

more text and comments

## Subheader 2

etc ...

Fragment 2:

```
```markdown
# Header of section 1

## Subheader 1

Some text

```{r}
some code
```

more text and comments

## Subheader 2

etc ...

Which one yields proper text?

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- If your document fails, note the reason. If it is due to ‘knitr’, it means your code has an error. Ch

processing file: ####.Rmd

Error in object[[i]]: ! object of type ‘closure’ is not subsettable Backtrace:

1. rmarkdown::render(...) ...
  17. base (local) docall(restart\$handler, restartArgs)
  18. evaluate (local) fun(base::quote(<ntSbsttE>))

Quitting from lines 17-20 [unnamed-chunk-2] (####.Rmd) Execution halted

Notice that in the example above, code from the source file ‘####.Rmd’ was not executed, and the interm

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If ‘knitr’ did its job, but the document still failed, it means your code is fine, but your text causes

|..... | 12% [unnamed-chunk-1] processing file: ####.Rmd “C:/Program Files/RStudio/resources/app/bin/quarto/bin/to  
and other pandoc stuff output file: ####.knit.md

! LaTeX Error: Unicode character ^B (U+0002) not set up for use with LaTeX. ... Error: LaTeX failed  
to compile ####.tex. See <https://yihui.org/tinytex/r/#debugging> for debugging tips. See ####.log for  
more info. Execution halted

Notice, that in this example, code in the original RMD document (‘processing file: ####.Rmd’) was execu

If you encounter this kind of problem, consider removing different chunks of text to locate the source c

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To save your time (if you need to knit your document many times and/or it contains some time-consuming s

```
‘‘‘r
knitr::opts_chunk$set(
 ... # different knitr options
 cache = TRUE
)
```

- Explain your workflow! For example, you are cleaning data, explain what you plan to do briefly and why you want to do it. Then, leave some comments about the data checks. The same about checking the assumptions for the the statistical approach you choose. It will allow us to follow your logic easier!

Also, discuss your results at least briefly. Although it may take time, it will help us to understand your ideas better => and your mark may be improved. If we explicitly tell you to present us the result of your analysis, we want you to give as the formal summary of your results with some statistics, p-value (but **not** only the p-value), possibly, an effect size estimate, and your conclusion. Consider the following link as an example of what we would like to see: **How to report t-test results** (by the way, this website has many useful information and hints!).

## Additional datasets

Here are some data sets for self-practice (can be called and attached by the `data(...)` command). These data sets are associated with some packages, so you need to have these packages too.

- **datasets** (built under the base R and provided by default):
  - **ToothGrowth**. Analyze the effect of the vitamin C supplementation form (**supp**: fresh orange juice – OJ or vitamin C synthetic formula – VC) and the dose (**dose**: 0.5, 1, 2 mg/ml) on the tooth length (**len**). (Already used in the mock exam paper for semester 2.)
  - **cars**. Analyze the effect of the car speed (**speed**) on the distance before full stopping (**dist**). Are these two variables linked? What is the relationship between them? (Already used in the semester 2 mock exam, question 1.)
  - **AirPassengers**. Decompose the provided time series into the trend, seasonality, and random variance. Does this model explain the data fully? Is there any unexplained variance left?
  - **InsectSprays**. Which insecticide is stronger?
  - **Loblolly**. Is there a link between the height of the tree and its age? How long would it take to get a 25-feet-high tree?
  - **PlantGrowth**. Which treatment increases the plant growth?
  - **trees**. What affects the volume of timber obtained from the cherry trees – the diameter (here, girth) of the tree trunk or the height of the tree? Why do you think so?
  - **iris**. Is there a difference in the sepal and petal lengths and widths of plants of different species? How would you classify your data?
- **datarium**:
  - **mice**. Is the average weight of the mice in this sample **equal to** 25 g? Is it **larger than** 20 g? Is it **smaller than** 18 g? Try different types of tests.
  - **mice2**. Is the weight *before* treatment **equal** to the one *after* the intervention? Is it larger *after* the treatment? Try different types of tests.
  - **genderweight**. Does the average body weight differ depending on the person's gender? Try different types of tests.
  - **properties**. This data set includes data about what type of property (house) was bought by what type of clients (singles or families). Is there any difference?
  - **titanic.raw**. The data shows which people survived the Titanic wreck and provides information about the age (**Child** or **Adult**), the status of the person (**Crew**, 1-4 **class**), and their sex (**Male** or **Female**). Are the chances of surviving the same across all different categories of passengers?
- **tidyverse**:
  - **relig\_income**. Is there a link between a person's religion and their personal income?
  - **diamonds**
  - **mpg**.
  - **starwars**. Among the characters of the Star Wars franchise, select only droids and humans. What is their distribution according to gender, species, and homeworld? What is the chance of selecting a feminine droid by chance?
  - **faithful**. Does the time between eruptions of the geyser affect the duration of the subsequent eruption?

In these data sets, load and inspect them, make a plot if suitable, and answer the question(-s) providing a clear solution. Also, clean the data if needed and convert it to metric units if different units are used in those data sets (convert miles, feet, or inches to the corresponding metric units). I have no solution for these data sets, but you can create them and send them to me for a brief discussion. After all, you have already seen this kind of question and you have an idea of what to expect.

More information about the respective data sets can be obtained by asking R to give help about the corresponding data set. For example, `?iris` will show information about this data set. If you feel that you would like to request more observations, you must remember that you need to provide us with the corresponding power analysis.

Good luck! Contact our course team if you have any questions.

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Originally, created by Dmytro Shytikov in 2025.